

BIOL B215: BIOSTATISTICS WITH R

7.1 Probability Models & Contingency Tables

Learning Goals

- Develop & test null hypotheses for count data.
- Know when to use the χ^2 (chi-square) distribution
- Calculate χ^2 from count data
- Diagnose if data meet χ^2 assumptions
- Know what to do when data do not meet χ^2 assumptions
- Use diverse distributions (Proportional, Binomial, Poisson etc.) to generate a null and test this null with a χ^2 goodness of fit test

Outline

- Goodness-of-fit tests
- The proportional model
- The χ^2 distribution
- Degrees of freedom
- The χ^2 test
- Using χ^2 to test if data are binomially distributed
- Using χ^2 to test if data are distributed according to a poisson distribution

GOODNESS-OF-FIT TESTS

Goodness-of-Fit Tests (1/3)

- Does data come from a given distribution with specified parameters?
- These tests compare **observed** frequency and **expected** frequency distributions under a probability model
- These tests use the χ^2 (chi-square) test statistic
- The test statistic measures the discrepancy between the observed frequency distribution and the frequencies expected under a simple probability model (which serves as the null hypothesis)

Goodness-of-Fit Tests (2/3)

- The **binomial** is one such probability model and the binomial test is a goodness-of-fit test
- Can we have a more general goodness-of-fit test that handles categorical and discrete variables with more than 2 outcomes?

The proportional model

- A simple **probability model** in which the frequency of occurrence of events is proportional to the number of opportunities

Example: NHL player birth months (1/)

Biological/scientific hypothesis:

Being born earlier in the year helps hockey players stand out when young, helping them achieve later success.

Example: NHL player birth months (2/)

How to test this?

What do we need?

What is the null?

What's the alternative?

H_0 :

H_A :

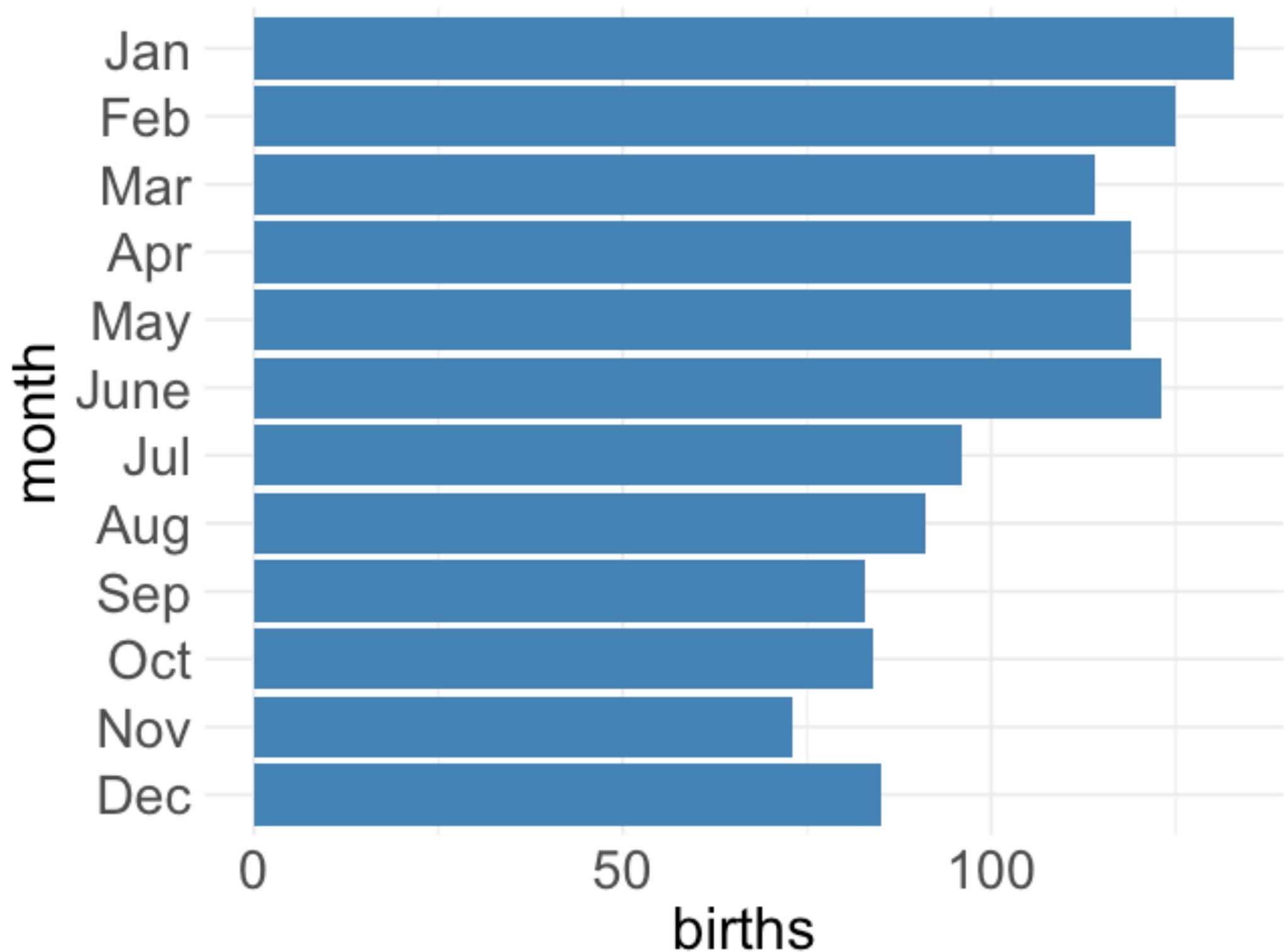
**Work on this for
a bit**

Example: NHL player birth months (3 /)

H_0 : Elite hockey players are born in random months.

H_A : Elite hockey players are NOT born randomly.

Example: NHL player birth months (4/)



Example: NHL player birth months (5/)

H_0 : The proportion of birth months of NHLers is like that of other people.

H_A : The proportion of birth months of NHLers is unlike that of other people.

Example: NHL player birth months (6/)

Human births by month:

Month	Jan	Feb	Mar	Apr	May	Jun
Proportion Births	.0794	.0763	.0872	.0863	.0895	.0876

Month	Jul	Aug	Sept	Oct	Nov	Dec
Proportion Births	.0854	.0850	.0854	.0819	.0770	.0787

Ventura, S. J., et al. 2001. Births: final data for 1999. *National Vital Statistics Reports* Vol. 49, No. 1.

Example: NHL player birth months (7/)

Introducing χ^2

χ^2 (chi-square) summarizes the (*lack of*) fit of **categorical data** to expectations.

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Example: NHL player birth months (8/)

Observed counts

Chi-square statistic

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Expected counts

The diagram illustrates the chi-square formula. A curved line connects the 'Observed counts' label to the O_i term in the numerator. Another curved line connects the 'Expected counts' label to the E_i term in the denominator. A third curved line connects the 'Chi-square statistic' label to the χ^2 symbol.

$\chi^2 = 0$ Data perfectly match null hypothesis expectations

$\chi^2 > 0$ Data show deviation from null hypothesis expectations

Example: NHL player birth months (9/)

Finding Expectations

Expected number of births in month i is:

$$E[X] = n_{total} \times \text{Null proportion}$$

$$n_{total} = 1245$$

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sept	Oct	Nov	Dec
Null Proportion	.0794	.0763	.0872	.0863	.0895	.0876	.0854	.0850	.0854	.0819	.0770	.0787
Expected # births	98.9	95.0	108.6	107.4	111.4	106.7	109.1	105.8	106.3	102.0	95.9	98.0
Observed # births (NHL)	133	125	114	119	119	123	96	91	83	84	73	85

Example: NHL player birth months (10/)

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Example, January:

$$\frac{(O_i - E_i)^2}{E_i} = ?$$

Work on this for a bit

Example: NHL player birth months (11/)

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Example, January:

$$\frac{(O_i - E_i)^2}{E_i} = \frac{[133 - (0.0794 \times 1245)]^2}{(0.0794 \times 1245)} \approx 11.8$$

Example: NHL player birth months (12/)

Month	Jan	Feb	Mar	Apr	May	Jun
Observed # births	133	125	114	119	119	123
Expected # births	98.9	95.0	108.6	107.4	111.4	106.7
$\frac{(Obs - Expect)^2}{Expect}$	11.8	9.5	0.3	1.2	0.5	2.5

Month	Jul	Aug	Sept	Oct	Nov	Dec
Observed # births	96	91	83	84	73	85
Expected # births	109.1	105.8	106.3	102.0	95.9	98.0
$\frac{(Obs - Expect)^2}{Expect}$	1.6	2.1	5.1	3.2	5.5	1.7

$$\chi^2 = 11.8 + 9.5 + 0.3 + 1.2 + 0.5 + 2.5 + 1.6 + 2.1 + 5.1 + 3.2 + 5.5 + 1.7$$

$$\chi^2 \approx 45$$

FINDING χ^2 IN R (I/)

```
#creating an nhl object based on actual data
>nhl <- tibble( month = c("Jan", "Feb",
"Mar", "Apr", "May", "June", "Jul", "Aug",
"Sep", "Oct", "Nov", "Dec"),
               month.num = 1:12,
               births = c(133, 125, 114, 119,
119, 123, 96, 91, 83, 84, 73, 85)) %>%
  mutate(month = fct_reorder(.f = month, .x =
month.num, .desc = TRUE))
```

FINDING χ^2 IN R (1/7)

#here's what it looks like

```
> print(nhl)
# A tibble: 12 × 3
  month month.num births
  <fct>      <int>   <dbl>
1 Jan         1    133
2 Feb         2    125
3 Mar         3    114
4 Apr         4    119
5 May         5    119
6 June        6    123
7 Jul         7     96
8 Aug         8     91
9 Sep         9     83
10 Oct        10     84
11 Nov        11     73
12 Dec        12     85
```

FINDING χ^2 IN R (2/7)

#human births. I created this object.

```
> human_births
# A tibble: 12 × 2
  month `births (%)`
  <chr>         <dbl>
1 Jan          7.94
2 Feb          7.63
3 Mar          8.72
4 April        8.63
5 May          8.95
6 June         8.57
7 Jul          8.76
8 Aug          8.5
9 Sept         8.54
10 Oct         8.19
11 Nov         7.7
```

FINDING χ^2 IN R (3/7)

```
#let's add an "expected" column
```

```
>nhl <- nhl %>%
```

```
  mutate(expected.prop = c(.0794, .0763, .0872, .0863, .0895, .0857,  
    .0876, .0850, .0854, .0819, .0770, .0787)) %>%
```

```
  mutate(expected.n = expected.prop * sum(births),  
    `expected %` = 100 * expected.prop)
```

```
> head(nhl)
```

```
# A tibble: 6 × 6
```

	month	month.num	births	expected.prop	expected.n	`expected %`
	<fct>	<int>	<dbl>	<dbl>	<dbl>	<dbl>
1	Jan	1	133	0.0794	98.9	7.94
2	Feb	2	125	0.0763	95.0	7.63
3	Mar	3	114	0.0872	109.	8.72
4	Apr	4	119	0.0863	107.	8.63
5	May	5	119	0.0895	111.	8.95
6	June	6	123	0.0857	107.	8.57

FINDING χ^2 IN R (4/7)

```
# calculate  $((O-E)^2)/E$   
nhl <- nhl %>%  
  mutate(` $((O-E)^2)/E$ ` = (expected.n -  
births)^2 / expected.n)
```


FINDING χ^2 IN R (5/7)

```
#let's see how it's looking
```

```
> nhl
```

```
# A tibble: 12 × 7
```

	month	month.num	births	expected.prop	expected.n	`expected %`	`((O-E)^2)/E`
	<fct>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	Jan	1	133	0.0794	98.9	7.94	11.8
2	Feb	2	125	0.0763	95.0	7.63	9.48
3	Mar	3	114	0.0872	109.	8.72	0.272
4	Apr	4	119	0.0863	107.	8.63	1.24
5	May	5	119	0.0895	111.	8.95	0.515
6	June	6	123	0.0857	107.	8.57	2.49
7	Jul	7	96	0.0876	109.	8.76	1.56
8	Aug	8	91	0.085	106.	8.5	2.08
9	Sep	9	83	0.0854	106.	8.54	5.12
10	Oct	10	84	0.0819	102.	8.19	3.17
11	Nov	11	73	0.077	95.9	7.7	5.45
12	Dec	12	85	0.0787	98.0	7.87	1.72

FINDING χ^2 IN R (6/7)

```
# calculate  $((O-E)^2)/E$ 
> nhl%>% select(month, births,
  expected.n,
  `((O-E)^2)/E` ) %>%
  mutate(expected.n = round(expected.n, digits = 1),
  `((O-E)^2)/E`=round(`((O-E)^2)/E`, digits = 1))
```

```
# A tibble: 12 × 4
```

	month	births	expected.n	$((O-E)^2)/E$
	<fct>	<dbl>	<dbl>	<dbl>
1	Jan	133	98.9	11.8
2	Feb	125	95	9.5
3	Mar	114	109.	0.3
4	Apr	119	107.	1.2
5	May	119	111.	0.5
6	June	123	107.	2.5
7	Jul	96	109.	1.6
8	Aug	91	106.	2.1
9	Sep	83	106.	5.1
10	Oct	84	102	3.2
11	Nov	73	95.9	5.5
12	Dec	85	98	1.7

FINDING χ^2 IN R (7/7)

```
#add the values in the  $((O-E)^2)/E$  column
```

```
>obs.chi2 <- summarise(.data = nh1,  
                        chi2 = sum(` $((O-E)^2)/E``))$ 
```

```
>obs.chi2  
# A tibble: 1 × 1  
  chi2  
  <dbl>  
1  44.9
```

From Test Stat (χ^2) to P-value (1/4)

By simulation:

- Randomly assign hockey players birth dates.
- Calculate χ^2 for these random data following the null.
- Repeat this many times.
- Compare observed χ^2 to χ^2 generated under the null.